

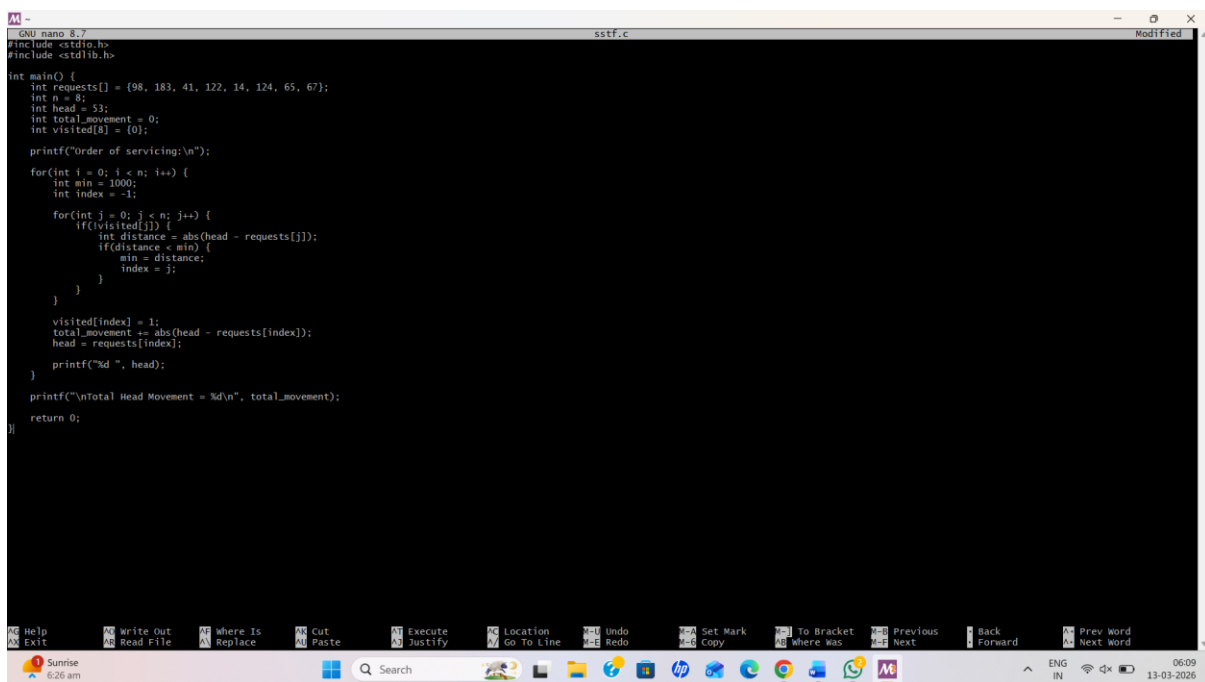
Practical No : 10

AIM :

Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests. Give me c code

Input :

```
bhavi@LAPTOP-IH23AUG3 MSYS ~  
$ nano sstf.c
```



```
GNU nano 8.7 sstf.c  
#include <stdio.h>  
#include <stdlib.h>  
  
int main() {  
    int requests[] = {98, 183, 41, 122, 14, 124, 65, 67};  
    int n = 8;  
    int head = 53;  
    int total_movement = 0;  
    int visited[] = {0};  
  
    printf("Order of servicing:\n");  
    for(int i = 0; i < n; i++) {  
        int min = 1000;  
        int index = -1;  
  
        for(int j = 0; j < n; j++) {  
            if(!visited[j]) {  
                int distance = abs(head - requests[j]);  
                if(distance < min) {  
                    min = distance;  
                    index = j;  
                }  
            }  
        }  
  
        visited[index] = 1;  
        total_movement += abs(head - requests[index]);  
        head = requests[index];  
        printf("%d ", head);  
    }  
  
    printf("\nTotal Head Movement = %d\n", total_movement);  
    return 0;  
}
```

Output :

```
bhavi@LAPTOP-IH23AUG3 MSYS ~  
$ ./sstf  
Order of servicing:  
41 65 67 98 122 124 183 14  
Total Head Movement = 323
```